

Lottery aversion and arbitrage risk

Combined factor-model portfolio test

Practitioner factor-model runner

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Abstract

Model. Lottery aversion and arbitrage risk, one long-short book from 2 sleeves (Lottery-Demand, ArbitrageRisk). 5857 symbols (full screened universe), 2005-09-01 to 2026-07-01 (251 months).

Earned. -1.60% annual arithmetic return (-2.52% compounded), 13.58% volatility, Sharpe -0.12, HAC $t=-0.53$.

Survives? vs SPY market benchmark: alpha 3.73%, $t=1.44$. vs VAL+MOM+BAB+QMJ: alpha -7.37%, $t=-2.94$. Conventional bar is $|t| \geq 2$.

Best swap. (Standard model, LotteryDemand for BAB) improves: Sharpe 0.11 against -0.06 (+0.16). **Verdict.** Reject for now: it clears neither control at $|t| \geq 2$.

Exploratory model diagnostic from a single configured run, not evidence from an untouched confirmatory sample.

1 Research objective

Test whether the configured model earns a useful next-month return and whether the benchmark or standard factors explain it. Signals are observed at month-end; returns are next-calendar-month simple returns.

Construction. Normalize within month, combine components within sleeves, then combine sleeves at the stock level using Lottery Demand 50%, Arbitrage Risk 50%. Construct one portfolio; apply beta neutrality and costs once. Sleeve returns are not averaged.

2 Configuration and sample

Factor	Composite sig-nals	Signal trans-form	Composite transform	Winsor	Min inputs	Mean inputs
Lottery Demand	lottery_max_5_21d, skew_63d	rank	rank	1%-99%	2	2.0
Arbitrage Risk	ivol_21d	rank	rank	1%-99%	1	1.0
Model con-struction	Book	Beta neutral	Bucket frac.	Cost bps	Min factors	Benchmark
signal weighted	long-short	no	0.33	0.0	2	SPY
Portfolio		Mean stocks		Minimum stocks		
Combined model		1716.8		20		
Lottery Demand		1930.2		23		
Arbitrage Risk		1716.9		20		

Sample: 5,857 symbols, 494,433 stock-months, 2001-07-01–2026-08-01. Screens: price 1.0, 21-day dollar volume None, market cap None, top-cap fraction 0.75, exclude financials False. Benchmark: SPY. Factor controls: standard_recreated_factors.csv.

3 Portfolio returns

Raw net performance on one shared window. Return and volatility use monthly arithmetic moments; t is HAC.

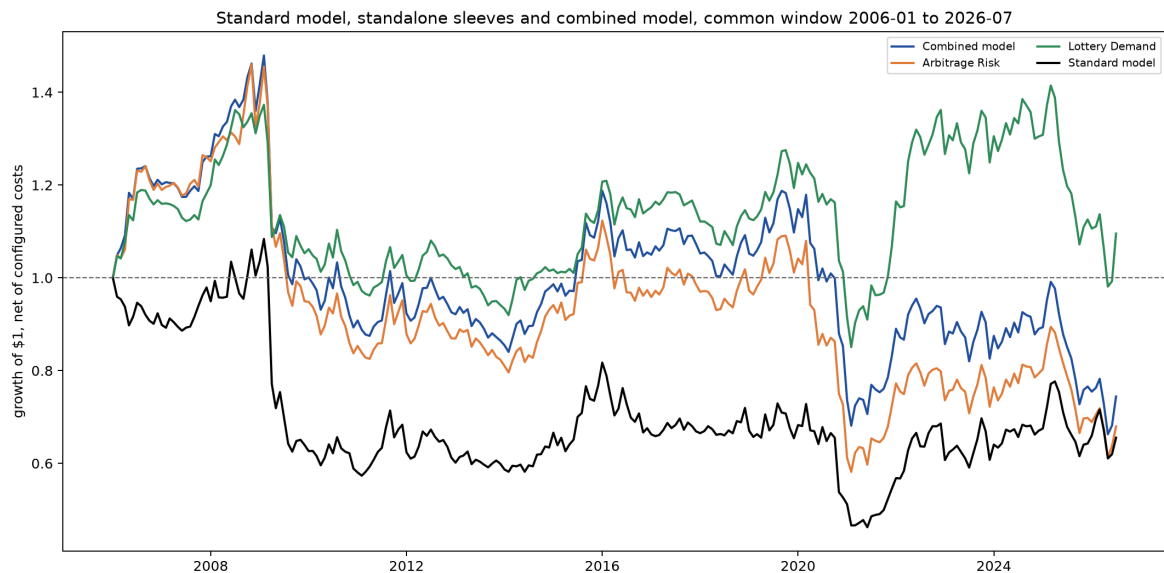


Figure 1: Growth of one dollar in each book, net of configured costs, on the common window. The path begins at one before the first return month.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	251	-1.60%	-2.52%	13.58%	-0.12	-0.53
Lottery Demand	251	0.04%	-0.64%	11.61%	0.00	0.01
Arbitrage Risk	251	-1.99%	-2.96%	14.05%	-0.14	-0.64
Standard model	246	-0.98%	-2.04%	14.42%	-0.07	-0.30

4 Benchmark fit and benchmark-orthogonalized returns

Regress each raw net return on SPY market benchmark. R^2 is variance explained; alpha is the annualized intercept; t and F use HAC inference. This is separate from standard-factor attribution. Estimated MKT loading: -0.457. Annual raw return -1.60% = controlled alpha 3.73% + benchmark contribution -5.33%. Orthogonal means zero sample covariance with the benchmark, not a lower mean. This is an ex-post regression diagnostic, not beta-neutral construction or a tradeable improvement.

Portfolio	N	Loading (MKT)	t (loading)	R^2 (SPY market benchmark)	F
Combined model	251	-0.46	-7.67	0.255	58.77
Arbitrage Risk	251	-0.48	-8.40	0.264	70.48
Lottery Demand	251	-0.34	-5.82	0.190	33.85
Standard model	246	-0.57	-8.01	0.356	64.22

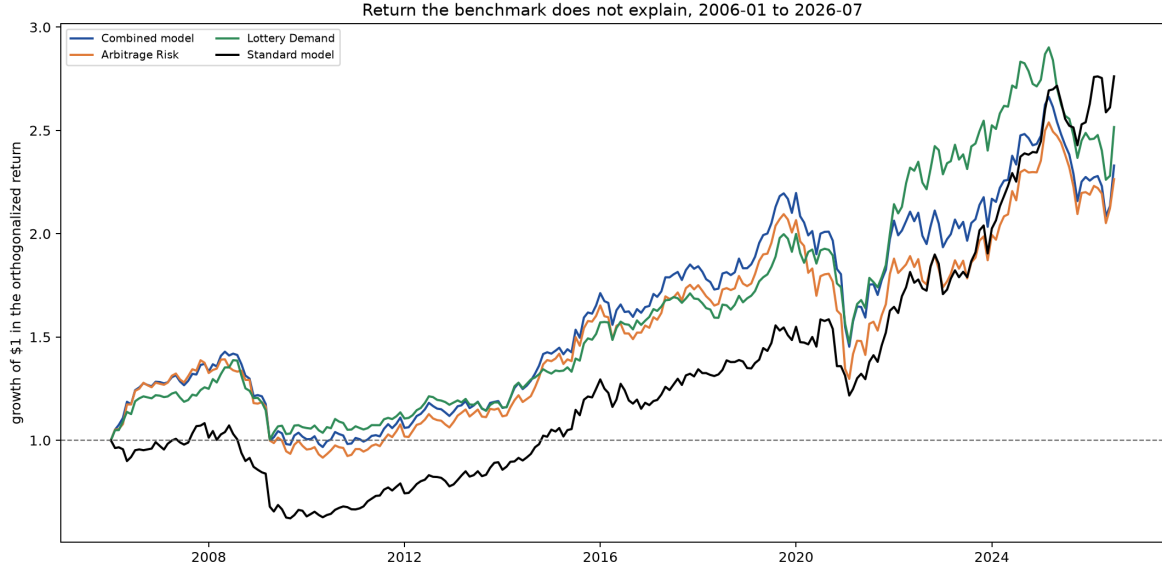


Figure 2: Returns orthogonalized to the SPY market benchmark: growth of one dollar in each book once the SPY market benchmark component is removed. A flat path earns nothing the benchmark does not already deliver.

The table summarizes the residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	251	3.73%	3.08%	11.72%	0.32	1.43
Arbitrage Risk	251	3.62%	2.92%	12.06%	0.30	1.34
Lottery Demand	251	3.97%	3.48%	10.45%	0.38	1.70
Standard model	246	5.65%	5.08%	11.57%	0.49	2.11

5 Attribution to configured factor controls

Regress each raw net return simultaneously on VAL+MOM+BAB+QMJ: $r_{p,t} = \alpha + \sum_k \beta_k f_{k,t} + \varepsilon_t$. The first table gives the loadings, with HAC t in brackets.

Portfolio	VAL	MOM	BAB	QMJ
Combined model	0.14 (1.92)	0.12 (1.80)	0.17 (1.75)	0.54 (7.52)
Arbitrage Risk	0.14 (1.73)	0.13 (1.96)	0.10 (1.00)	0.59 (7.09)
Lottery Demand	0.22 (3.10)	0.07 (1.08)	0.18 (2.28)	0.43 (7.05)

Return contribution is $\beta_k \times 12 \times \overline{f_k}$. Explained plus alpha equals total annual return.

Portfolio	VAL	MOM	BAB	QMJ	Explained	Total
Combined model	-0.16%	0.67%	0.55%	4.25%	5.30%	-2.07%
Arbitrage Risk	-0.15%	0.70%	0.32%	4.61%	5.48%	-2.34%
Lottery Demand	-0.24%	0.39%	0.58%	3.33%	4.06%	-0.18%

5.1 Simultaneous factor spanning and factor-orthogonalized returns

The same regression as a spanning test. R2 is the share of monthly net-return variance jointly explained by the factor controls.

Portfolio	N	R2 (4F)	F
Combined model	230	0.452	29.44
Arbitrage Risk	230	0.434	27.48
Lottery Demand	230	0.446	24.37

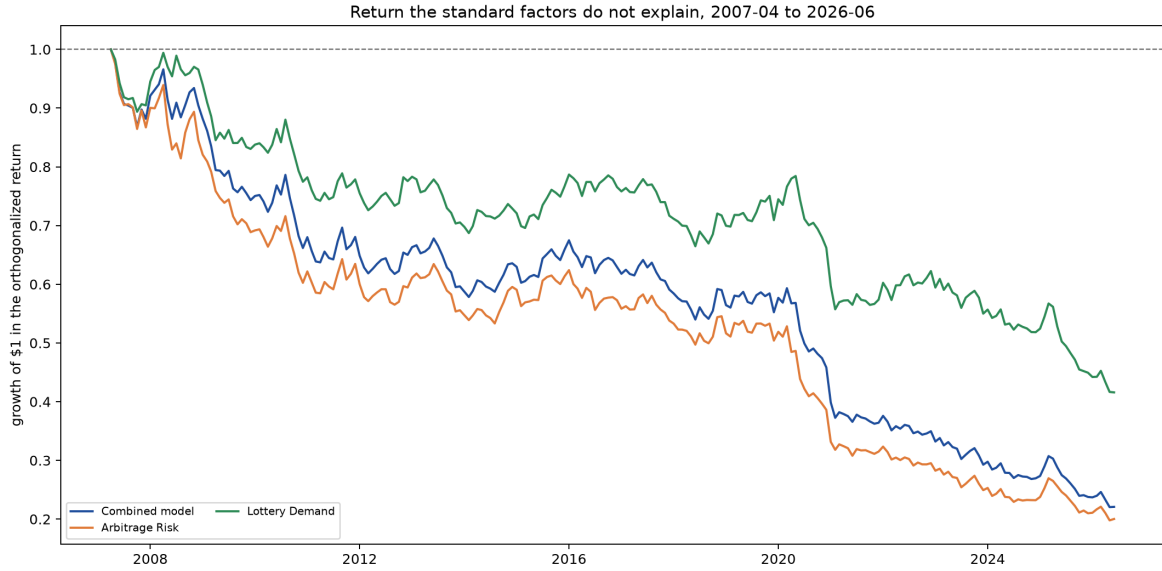


Figure 3: Returns orthogonalized to VAL+MOM+BAB+QMJ: growth of one dollar in each book once its factor exposures are removed. This is the Alpha column above, drawn as a path.

The table summarizes the factor-residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	230	-7.37%	-7.58%	9.75%	-0.76	-3.25
Arbitrage Risk	230	-7.82%	-8.05%	10.39%	-0.75	-3.29
Lottery Demand	230	-4.23%	-4.47%	8.18%	-0.52	-2.16

5.2 The new model against the reference model

Regress the combined model on Standard model as one return series. High R2 with zero alpha means the reference reproduces the model.

Portfolio	N	Alpha	t	R2 (Standard model)	F
Combined model	246	0.20%	0.12	0.702	528.91

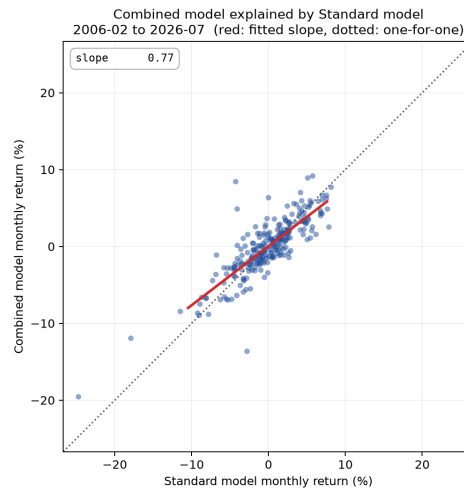


Figure 4: Each month of the combined model against Standard model, with the fitted slope. A tight cloud along the line is a model the reference reproduces; vertical spread is what it cannot.

5.3 Correlation and principal components

Check whether a sleeve duplicates a standard factor. The combined model is excluded because its overlap with its own sleeves is mechanical. PCA uses standardized sleeve and standard-factor returns over 230 common months. The share shown in the plot is the largest covariance-matrix eigenvalue divided by the sum of all eigenvalues: the fraction captured by one dominant co-movement, not return or regression R2. The combined model is excluded because its overlap with its own sleeves is mechanical.

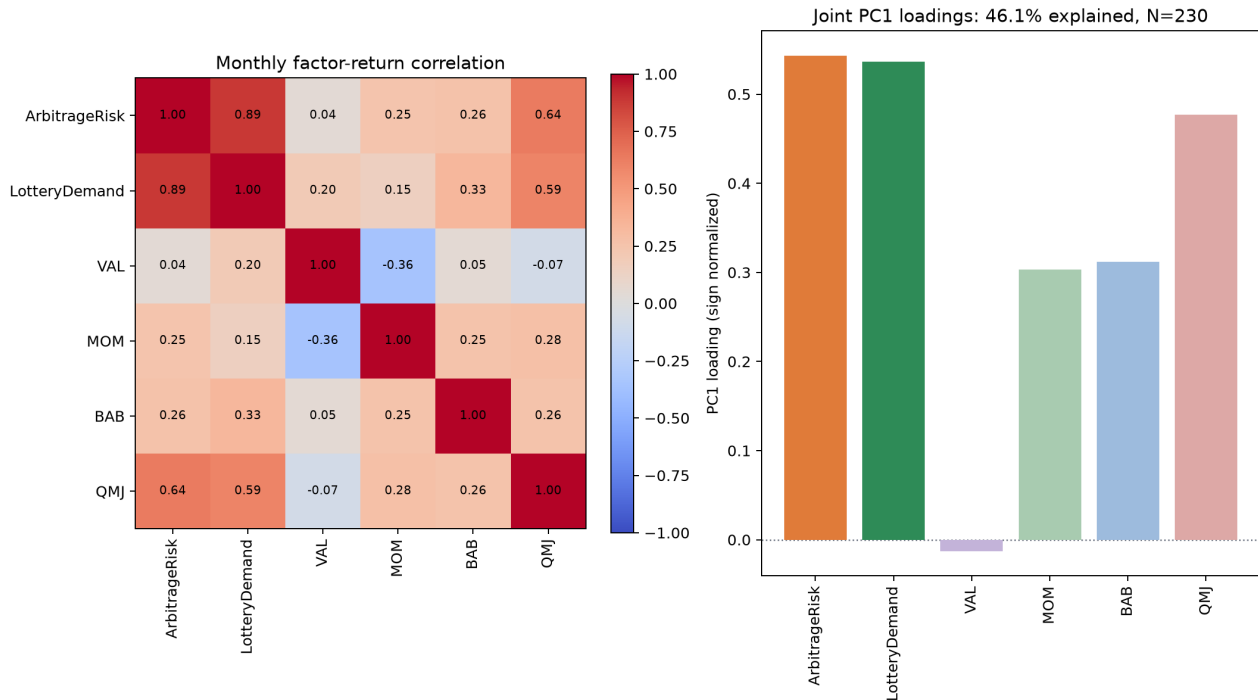


Figure 5: Monthly return correlations and sign-normalized first-principal-component loadings across the standalone sleeves and the supplied standard factors.

6 Signal persistence

Mean rank IC between formation exposure and a single later month's return. Flat profiles persist; fast decay implies faster rebalancing.

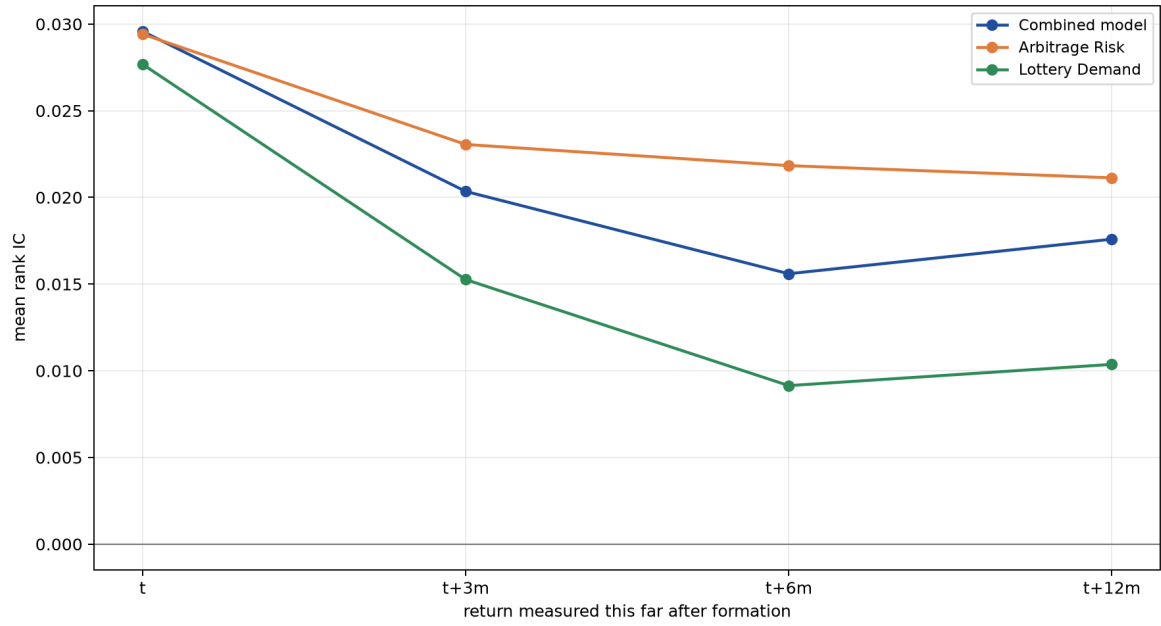


Figure 6: Mean rank information coefficient at each horizon. A line that stays flat is a signal that keeps working long after the month it is traded.

Portfolio	Horizon	Mean IC	t	Months
Combined model	t	0.0296	3.42	251
Combined model	t+3m	0.0203	2.06	248
Combined model	t+6m	0.0156	1.70	244
Combined model	t+12m	0.0176	2.17	237
Arbitrage Risk	t	0.0294	3.30	251
Arbitrage Risk	t+3m	0.0231	2.33	248
Arbitrage Risk	t+6m	0.0218	2.24	244
Arbitrage Risk	t+12m	0.0211	2.57	237
Lottery Demand	t	0.0277	3.78	252
Lottery Demand	t+3m	0.0153	1.90	248
Lottery Demand	t+6m	0.0091	1.29	245
Lottery Demand	t+12m	0.0104	1.61	237

7 Candidate comparisons and replacement

Standalone compares candidate with incumbent. In-model replaces the incumbent sleeve and rebuilds the standard model; diversification can reverse the standalone ranking.

Each pair judged twice: as a standalone factor, and swapped into the model
dashed line: the unmodified standard model (-0.84%)

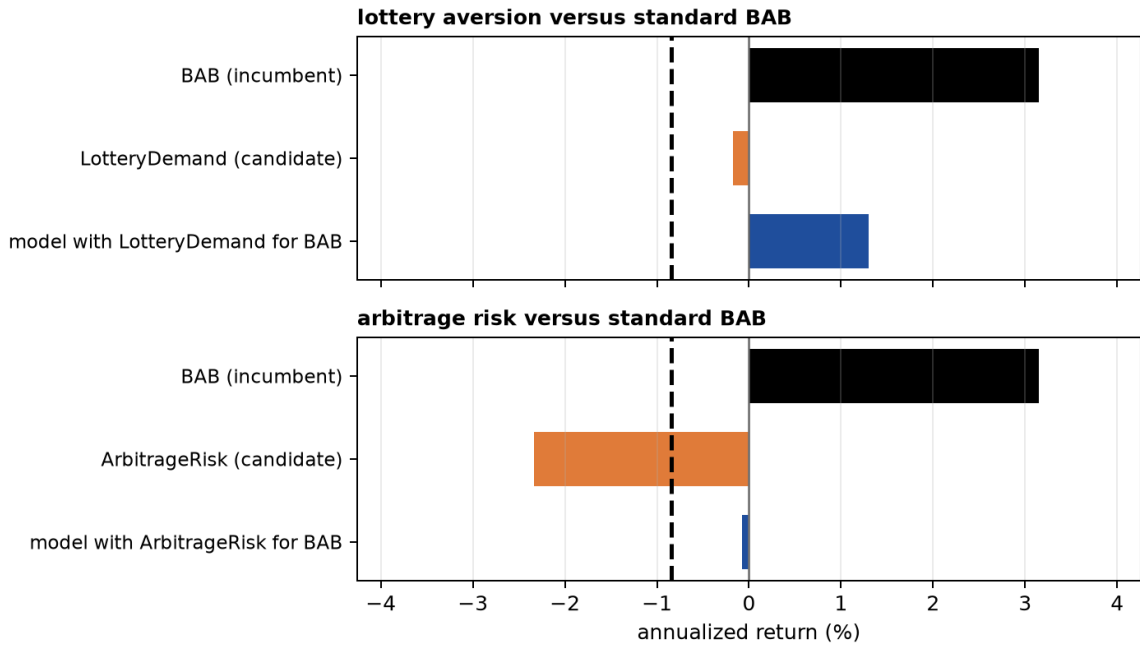


Figure 7: Each pair judged twice: the incumbent factor against the candidate standing alone, then the model with the candidate swapped in. The dashed line is the unmodified standard model.

Pair	Role	Portfolio	Ann. ret.	Change	Alpha	Change	t
lottery aversion versus standard BAB	incumbent	BAB	3.15%	—	3.22%	—	1.03
	candidate in model	Lottery Demand	-0.18%	-3.33%	3.61%	+0.38%	1.50
		Standard model, Lottery Demand for BAB	1.30%	+2.14%	5.35%	-0.44%	1.89
arbitrage risk versus standard BAB	incumbent	BAB	3.15%	—	3.22%	—	1.03
	candidate in model	Arbitrage Risk	-2.34%	-5.49%	3.06%	-0.17%	1.18
		Standard model, Arbitrage Risk for BAB	-0.08%	+0.76%	4.75%	-1.04%	1.79

8 Conclusion

Verdict. Reject for now: it clears neither control at $|t| \geq 2$. vs SPY market benchmark: alpha 3.73%, $t=1.44$; vs VAL+MOM+BAB+QMJ: alpha -7.37%, $t=-2.94$.

9 Reproducibility

Run ID:

20260826T141021Z_lottery-aversion-and-arbitrage-risk_6d0bf0259e

Configuration hash: 6d0bf0259e. Gold version: version=20260826T131322Z. The run directory contains configuration, data products, figures, source, log, and PDF. Re-runs never overwrite it.